Qualifying Exam - Data Analysis

Identifier Number: 8768

June 20, 2025

Contents

"I attest that upon completion of this exam, I have destroyed the data file and I acknowledge that these data should not be used for any other purposes."

1 Abstract

Physical inactivity increases risks for chronic diseases like heart disease and diabetes, motivating widespread use of mobile apps and text messages to promote walking. However, it remains unclear whether customizing these messages based on a person's activity level yields better results than sending generic reminders. To address this, we conducted a 14-day randomized trial with 298 adults who wore step-counting devices. Participants were randomly assigned to one of three groups: no messages (Control), daily generic motivational messages (Standard), or daily messages tailored using the participant's collected data (Tailored). During the first week, participants receiving generic and tailored messages increased their daily step counts by approximately 52% (95% CI: 40–66%) and 57% (44–71%), respectively, compared to the Control group. These increases translated into roughly 630–2,000 additional daily steps, depending on initial activity levels ($p < 0.001$). Both messaging groups showed a similar pattern of activity over the full two weeks, peaking around Day 7 (approximately 1,300 steps per day above Control), followed by a gradual decline. Surprisingly, tailored messages offered no significant advantage over generic prompts. Further, a predictive model based on baseline factors and first-week activity accurately explained 87% of participants' second-week steps. Notably, both messaging strategies increased variability between individuals with some participants improved dramatically, while others showed little change. These findings suggest that simple, well-timed generic reminders effectively boost short-term physical activity, especially for less active individuals. However, the substantial variability in individual responses highlights the need for adaptive interventions targeting low responders. Limitations include the study's brief duration, the use of participants who may already be motivated, and the difference in time where the user wears the devices, which might not generalize to other populations or longer-term outcomes.

2 Introduction

Regular physical activity protects against cardiovascular disease, type 2 diabetes, certain cancers, and premature death while improving mental health and quality of life. Despite these well documented benefits, a majority of American adults fail to meet recommended physical activity guidelines, making sedentary lifestyle one of the leading preventable causes of disease and healthcare costs.

The widespread adoption of smartphones and wearable fitness devices has created opportunities to deliver behavior changing interventions directly to individuals throughout their daily lives which can passively track activity and automatically send motivational messages at strategic times. Mobile health (mHealth) interventions have become increasingly popular, with millions of people using fitness apps that help them become more active.

Among these digital tools, motivational text messages represent one of the simplest and most scalable approaches. Most fitness apps send some form of encouraging reminders, ranging from basic prompts like "Don't forget your daily walk!" to more sophisticated messages that reference users' recent activity patterns or personal goals. The appeal of message-based interventions lies in their low cost, broad reach, and ability to provide support precisely when people might need encouragement to be active.

However, whether or not personalizing messages actually improves their effectiveness remains unclear. This uncertainty has important practical implications because if generic messages work just as well as personalized ones, intervention designers could save considerable resources while achieving similar outcomes. Understanding what works best could help influence the health behaviors of the users tremendously.

To address this question, a controlled experiment was conducted comparing three approaches: no messages, generic daily messages, and messages personalized based on each participant's recent activity data and baseline characteristics. The study tracked participants' daily step counts over two weeks using wearable devices, allowing the examination of both immediate effects and changes over time. The study was specifically designed to answer four key questions about how motivational messaging affects physical activity behavior:

- First Week Comparison: How do the three intervention groups compare in average daily step count during the first week? How does this comparison vary by baseline activity level?

- 14-Day Trajectories: How do step count trajectories evolve over the full 14-day period, and are these patterns different across intervention groups?
- Prediction: Can participants' Week 2 activity be accurately predicted using their baseline covariates and step counts from Days 1–7?
- Individual Variability: To what extent do individuals differ in their step count patterns, and is the magnitude of this variability different across intervention groups?

These questions address both the practical concerns of intervention designers and the scientific understanding of behavior change processes. By examining not just average effects but also individual variability and patterns of change over time, we are able to provide insights that could inform the development of more effective digital health interventions. The following sections detail our methods for addressing these questions and present findings that have important implications for the millions of people who rely on digital tools to support their health goals.

3 Methods

3.1 Source of the Data

The data come from a two-week randomized trial designed to evaluate the effectiveness of motivational messaging in promoting physical activity. Participants were randomly assigned to one of three intervention arms: Control (no messages), Standard (daily generic motivational messages), or Tailored (daily messages personalized based on prior behavior and participant characteristics). The exact recruitment methods and eligibility criteria for participants are not specified in the data documentation, but all enrolled individuals completed a baseline survey and were expected to wear a step-counting device throughout the 14-day study period.

The data was collected in long format, with each row representing a single day of observation for a single participant, resulting in a repeated measures structure. Each participant could contribute up to 14 observations, one for each study day, although the presence of missing data and non-wear days (coded as zero step counts) reduced the total number of usable data points.

Available variables include:

- id: Unique participant identifier (300 distinct values)
- day: Study day (day 1 through 14: starting on Monday and ending on the second Sunday)
- stepcount: Total steps recorded for that day (primary outcome with an average of 3,223 steps per day and an overall right skew)
- Group: Messaging intervention group assignment (Control, Standard, or Tailored (all 33.3%))
- Act: Baseline self-reported activity level (not active (51.7%), somewhat active (29.7%), very active (18.7%))
- Age: Age in years at enrollment (Average age of 27.9 years old with a range from 20 to 40 years old)
- Female: Binary sex indicator (1 = female (60.2%), 0 = male (39.8%))

3.1.1 Missing Data and Data Integrity

We assume missing data occurred from participants forgetting to wear their devices or device malfunctions, resulting in days with zero recorded steps. We treated these zeros as missing data rather than true inactivity, since people are unlikely to take no steps in a day. There were other low values of steps but due to uncertainty of people's lives or conditions we assumed these as valid. After this cleaning, about 4.7% of our total observations (198 out of 4,200) were missing step count data. ***Table X***. Step count missingness was generally scattered across time and groups without strong clustering, supporting the assumption that data is Missing at Random (MAR) ***Table X***. Two participants had no valid step data for the entire study period and were removed from the analysis, leaving us with 296 participants. A few participants (n=6) were also missing sex information from their baseline survey ***Table X***.

To address missing values, we used multiple imputation via predictive mean matching with the mice package in R. This method creates several complete datasets by predicting missing values based on observed patterns in the data, then combines results across these datasets to account for uncertainty about the missing values. This approach is generally more reliable than simply ignoring missing data, which can introduce bias if missingness is related to the outcomes we're studying. The validations can be seen in ***Table X***.

3.1.2 Potential Confounding and Precision Values

Randomization was conducted at baseline so measured and unmeasured covariates should be balanced across intervention groups on average. However, due to the modest sample size, residual imbalances are possible. We therefore adjusted all inferential models for baseline characteristics—age, sex, and baseline activity level (Act) to improve precision and reduce potential bias. Age and baseline activity likely affect overall physical activity capacity. Unmeasured factors that might influence daily activity such as occupation, weather, other responsibilities, or health conditions were not captured in our data. While randomization should distribute these factors evenly across groups, they represent a source of variability in our results.

3.2 Statistical Methods

All of the analysis was conducted in R 4.3.2 using a R markdown. Data reshaping and cleaning were performed using dplyr and tidyr. Visualizations were created using ggplot2, and formatted output tables were rendered using kableExtra. Missing data was addressed through multiple imputation using mice with five imputations. The multiple imputation diagnosis can be seen in ***Table X*** which validates its usage.

To evaluate intervention effects, we used linear regression models for cross-sectional outcomes (e.g., Week 1 averages), and linear mixed-effects models (lmer, from the lme4 package) for longitudinal analyses of daily step counts. These models included random intercepts and slopes to account for within-subject correlation and patterns in data over time. Model coefficients and confidence intervals were extracted using broom.mixed, and group-level contrasts were obtained via estimated marginal means (emmeans). Interaction terms (e.g., Group $\times$ Act, Group $\times$ Day) were tested to assess effect modification.

Nonlinear time effects were modeled using quadratic polynomial terms for day. Alternative spline models were considered, but diagnostics indicated that quadratic trends adequately modeled the observed trajectories. Model comparisons were conducted using Wald tests and likelihood ratio tests where appropriate. For prediction tasks (Question 3), we fit pooled linear models to predict Week 2 step counts using baseline covariates and observed Week 1 behavior. Model performance was evaluated using R^2 and predicted vs observed plots. Residual diagnostics (e.g., Q–Q plots, residuals vs fitted plots) were used to assess linearity, homoscedasticity, and normality assumptions.

Variance components from the mixed-effects model were used to quantify individual variability in activity patterns, and Levene's test was applied to compare within group variability in residual step counts. All plots and tables were constructed to highlight meaningful comparisons across intervention arms.

3.2.1 Scientific Question 1: First Week Comparison

We used linear regression with log-transformed step count + 1 (for possible step values of 0 after multiple imputation) as the dependent variable to compare daily step counts among the three intervention groups (Control, Standard, Tailored) during the first week. Log transformation was applied to address heteroscedasticity and non-normality observed in the untransformed data, as confirmed by residual diagnostics ***FIgure X***. The model included main effects for intervention group and baseline activity level (Not Active, Somewhat Active, Very Active), plus interaction terms to assess whether intervention effects varied by baseline activity.

$$log(Y_{ij} + 1) = \beta_0 + \beta_1 Group_i + \beta_2 Act_i + \beta_3 (Group_i * Act_i) + \epsilon_{ij}$$

The model was fit separately to each of the five multiply imputed datasets using the mice and with() functions in R, and pooled using Rubin's rules via the pool() function. Residual plots and QQ plots were examined to assess model assumptions, and the log-transformed model provided better residual behavior than the untransformed version.

This modeling approach allows for direct interpretation of group differences in daily activity levels, and the inclusion of interaction terms enabled us to test whether the effectiveness of the intervention varied by baseline activity level. We retained the model on the original (untransformed) scale for interpretability, as diagnostics did not suggest serious violations of linear model assumptions. A log-transformed version was evaluated for robustness but yielded similar conclusions. To interpret pairwise differences among intervention groups clearly and accurately, we performed post-hoc pairwise comparisons using Tukey's Honest Significant Difference (HSD) procedure, adjusting for multiple comparisons.

3.2.2 Scientific Question 2: 14-Day Step Count Trajectories

To evaluate how daily step counts evolved across the full 14-day study and whether this trajectory differed between intervention arms, we fit a linear mixed-effects model with subject-specific random intercepts and slopes. The outcome variable was the log-transformed step count (log(stepcount + 1)), consistent with the analysis for Question 1. The model included fixed effects for intervention group, a quadratic polynomial in study day, their interaction, and baseline covariates (age, sex, activity level). A Wald test comparing the full model to one excluding baseline covariates confirmed that age, sex, and activity level added explanatory value. The quadratic term was added to capture potential non-linear changes and due to the visual parabolic shape of the mean steps over time graph (***FIGURE X***). The model equation was:

$$\begin{aligned} log(Y_{ij} + 1) = \beta_0 + \beta_1 \text{Group}_i + \beta_2 \text{poly}(Day_j, 2) + \beta_3 (\text{Group}_i \times \text{poly}(Day_j, 2)) \\ + \beta_4 \text{Act}_i + \beta_5 \text{Age}_i + \beta_6 \text{Female}_i + b_{0i} + b_{1i} \cdot Day_j + b_{2i} \cdot Day_j^2 + \varepsilon_{ij} \end{aligned}$$

To account for repeated measurements within individuals, the model included random intercepts and random slopes for both the linear and quadratic terms of day for each participant $b_{0i}, b_{1i}, b_{2i} \sim N(0, G)$. This specification allows each individual to have their own unique step count trajectory We assess the assumptions of the final mixed-effects model by examining the residuals and random effects from a model fit on a single (the first) completed dataset. We check for normality and homogeneity of residuals, and for the normality of the random effects.

3.2.3 Scientific Question 3: Predicting Week 2 Step Counts

To evaluate how well baseline characteristics and first week step behavior predict future activity, we built a multiple linear regression model to forecast each participant's average daily step count during Week 2 (Days 8–14). The predictors included baseline covariates (intervention group, baseline activity level, age, and sex) and the individual step counts recorded on Days 1 through 7. The outcome variable was defined as the mean of the observed and imputed step counts from Days 8–14, while predictors consisted of the full set of baseline variables and step counts for Days 1–7. We fit the following model to each imputed dataset and then pooled the results using Rubin's rules:

$$\text{Week2Avg}_i = \beta_0 + \beta_1 \text{Group}_i + \beta_2 \text{Act}_i + \beta_3 \text{Age}_i + \beta_4 \text{Female}_i + \sum_{d=1}^{7} \beta_{4+d} \cdot \text{Step}_{id} + \epsilon_i$$

This modeling framework allows us to interpret each coefficient as the marginal association between a given predictor and the average Week 2 step count, holding all other variables constant. We assessed the fit of the model using residual plots from the first imputed dataset, which showed reasonable adherence to linear model assumptions. The residuals appear to have approximately constant variance and normal distribution, although a small number of influential points were observed. Additionally, multicollinearity diagnostics revealed moderate inflation among the week 1 step count predictors (e.g., VIFs ranging up to ~4.5), which was expected due to natural autocorrelation across consecutive days. However, none exceeded commonly used thresholds for concern (e.g., VIF $>$ 5), supporting the overall robustness of the model.

3.2.4 Scientific Question 4: Individual Variability

To quantify how much individuals differed in their activity patterns and whether the magnitude of this variability varied across groups, we examined the variance components from the mixed-effects model used in Question 2. These included the random intercept variance (baseline step count), and random slope variances for both the linear and quadratic time effects. Letting b_{0i}, b_{1i}, b_{2i} denote the subject-level random effects, the total between-person variability in trajectories was captured by $\text{Var}(b_{0i})$, $\text{Var}(b_{1i})$, $\text{Var}(b_{2i})$ Variance components were extracted from each imputed model and averaged to provide stable estimates. The pooled intercept and residual variances were used to compute the intraclass correlation coefficient (ICC)

$$ICC = \frac{\sigma^2_{intercept}}{\sigma^2_{intercept} + \sigma^2_{residual}}$$

To assess whether day-to-day variability differed by group, we conducted Levene's Test on the residuals from the model fit on the first imputed dataset. This test compares the dispersion of residuals across intervention arms and is robust to non-normality. A significant result indicated that the Standard and Tailored groups exhibited greater residual variability, suggesting more fluctuation in daily behavior. Residual distributions were visualized using boxplots to further illustrate the differences. These analyses support the interpretation that the intervention arms induced greater heterogeneity in individual responses, potentially due to variation in engagement with motivational messaging.

4 Results

4.1 Scientific Question 1: First Week Comparison

Both intervention approaches demonstrated clear improvement over control conditions during the first week. Relative to the Control arm, participants receiving Standard messages accumulated 52% more steps per day (estimate = 1.52; 95% CI 1.40 to 1.66; $p < 0.001$), while those receiving Tailored messages achieved a 57% increase (estimate = 1.57; 95% CI 1.44 to 1.71; $p < 0.001$). However, no statistically significant difference emerged between Standard and Tailored messages (all pairwise p-values   0.13), indicating that tailoring offered no clear advantage over generic messaging during this initial period. This was also seen in the pairwise test of the log(stepcount + 1) model given only Group. Independent of intervention type, baseline activity was strongly predictive of observed behavior. Compared with "not active" participants, "somewhat active" and "very active" participants recorded roughly 2.5 and 4 times higher daily steps, respectively (both $p < 0.001$).

Tukey's pairwise comparisons revealed that intervention effectiveness varied by baseline activity level. Among not active participants, both interventions produced similar slight gains (Standard: +627 steps/day; Tailored: +688 steps/day; between-group difference p = 0.730). The somewhat active group demonstrated the largest absolute benefits, with Standard exceeding Control by 2,023 steps per day ($p < 0.001$) and Tailored exceeding Control by 1,507 steps per day ($p < 0.001$), though the 515 step advantage for Standard was not significant (p = 0.131). Among very active participants, significant interactions emerged (Standard×very active: β = -0.226, p = 0.004; Tailored×very active: β = -0.193, p = 0.029), indicating diminished intervention effects. While both interventions remained effective compared to control (Standard: +1,029 steps per day, p = 0.011; Tailored: +1,420 steps per day, p = 0.002), the incremental benefit was 18-22% smaller than in the least-active group. This shows the importance of consistent messaging but less importance of tailoring messages.

Table 1: Pooled Results: Log-Transformed Model Week 1 Step Count

Covariate	Estimate	95% Confidence Interval	P-Value
(Intercept)	7.097	7.039 to 7.154	<0.001
GroupStandard	0.418	0.333 to 0.504	<0.001
GroupTailored	0.451	0.363 to 0.538	<0.001
Actsomewhat_active	0.924	0.827 to 1.020	<0.001
Actvery_active	1.388	1.274 to 1.501	<0.001
GroupStandard:Actsomewhat_active	0.092	-0.047 to 0.230	0.196
GroupTailored:Actsomewhat_active	-0.048	-0.186 to 0.090	0.493
GroupStandard:Actvery_active	-0.226	-0.381 to -0.070	0.004
GroupTailored:Actvery_active	-0.193	-0.366 to -0.020	0.029

Table 2: Pairwise Group Comparisons by Baseline Activity (in Steps/Day)

Comparison	Estimate	95% Confidence Interval	P-Value
Baseline: Not Active			
Control - Standard	-627.3	-788.3 to -466.4	<0.001
Control - Tailored	-687.6	-855.8 to -519.3	<0.001
Standard - Tailored	-60.2	-248.4 to 127.9	0.730
Baseline: Somewhat Active			
Control - Standard	-2022.7	-2581.9 to -1463.5	<0.001
Control - Tailored	-1507.4	-2006.2 to -1008.6	<0.001
Standard - Tailored	515.3	-111.8 to 1142.5	0.131
Baseline: Very Active			
Control - Standard	-1028.6	-1867.9 to -189.3	0.011
Control - Tailored	-1420.2	-2407.1 to -433.3	0.002
Standard - Tailored	-391.6	-1409.3 to 626.1	0.637

4.2 Scientific Question 2: 14-Day Step Count Trajectories

The 14-day step count trajectories revealed distinct patterns across the three intervention groups ***(Image 1).*** Both Standard and Tailored groups demonstrated similar quadratic patterns, with step counts rising from baseline levels around 3,000 steps per day to peak levels of approximately 4,300 steps per day around days 5-7, followed by a gradual decline toward the end of the study period. In contrast, the Control group maintained relatively stable step counts throughout the 14-day period, averaging approximately 2,500 steps per day with minimal variation over time. The confidence intervals show greater variability in the intervention groups compared to the control group, particularly during the peak activity period.

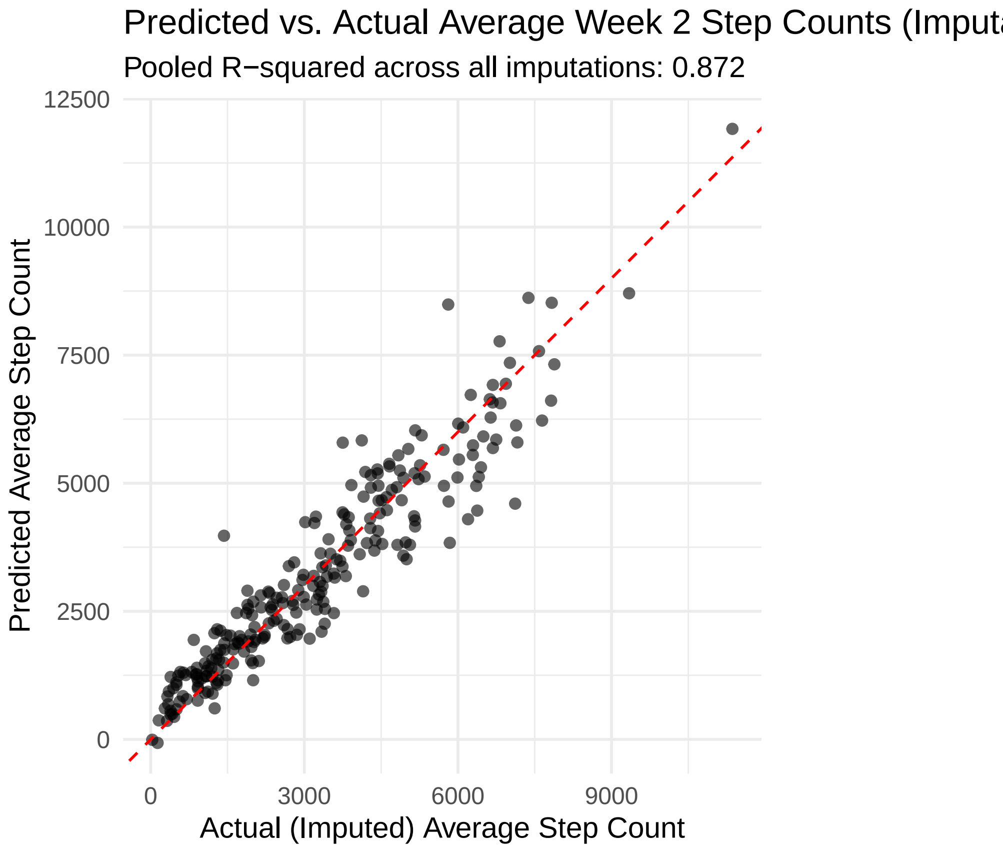
The mixed-effects model analysis confirmed significant main effects for both interventions compared to control over the 14-day period (Standard: 0.301 increase in log steps, 95% CI: 0.171 - 0.431, $p < 0.001$; Tailored: 0.273 increase, 95% CI: 0.142 - 0.404, $p < 0.001$). The polynomial time terms were not significant as main effects ($p = 0.734$ and $p = 0.739$), indicating no overall quadratic trend across all groups. However, highly significant interactions emerged between group assignment and the quadratic time components. Both Standard and Tailored groups showed significant negative interactions with the linear time component (Standard: -5.431, $p = 0.009$; Tailored: -6.683, $p = 0.001$) and more pronounced negative interactions with the quadratic component (Standard: -9.800, $p < 0.001$; Tailored: -7.324, $p < 0.001$).

The significant time interactions indicate that while both intervention groups maintained higher average step counts than control throughout the study, their activity patterns followed a concave trajectory rather than a constant linear increases. The interventions appeared most effective during the middle portion of the study period, with both groups achieving peak effectiveness around days 5-7 before showing evidence of declining. The Standard and Tailored interventions performed similarly overall, with the Standard group showing a slightly larger main effect but both groups displaying comparable trajectory patterns.

Table 3: Pooled Results for Mixed-Effects Model of 14-Day Step Trajectories

Covariate	Estimate	95% Confidence Interval	P-Value
(Intercept)	8.129	7.858 to 8.401	<0.001
GroupStandard	0.301	0.171 to 0.431	<0.001
GroupTailored	0.273	0.142 to 0.404	<0.001
poly(day, 2)1	-0.489	-3.316 to 2.338	0.734
poly(day, 2)2	-0.337	-2.318 to 1.644	0.739
Actsomewhat_active	0.953	0.841 to 1.064	<0.001
Actvery_active	1.229	1.101 to 1.358	<0.001
Age	-0.034	-0.042 to -0.025	<0.001
Female	-0.185	-0.285 to -0.085	<0.001
GroupStandard:poly(day, 2)1	-5.431	-9.512 to -1.351	0.009
GroupTailored:poly(day, 2)1	-6.683	-10.789 to -2.577	0.001
GroupStandard:poly(day, 2)2	-9.800	-12.854 to -6.745	<0.001
GroupTailored:poly(day, 2)2	-7.324	-10.414 to -4.234	<0.001

4.3 Scientific Question 3: Predicting Week 2 Step Counts



The predictive model using baseline covariates and first week daily step counts achieved strong performance in predicting average Week 2 step counts, with a pooled R-squared of 0.872 across all imputations meaning 87% of the variance in the dependent variable is explained by the independent variable. The predicted versus actual plot demonstrates good calibration across the full range of step counts, from sedentary participants averaging under 1,000 steps per day to highly active participants exceeding 10,000 steps per day. The strong linear relationship between predicted and actual values indicates that participants' first week activity patterns, combined with baseline characteristics, provide reliable predictors of subsequent activity levels during the second week of the study.

Counter to expectations from the trajectory analysis, both Standard and Tailored intervention groups showed significant negative associations with Week 2 step counts when controlling for first-week activity (Standard: -376 steps per day, 95% CI: -727 to -24, $p = 0.037$; Tailored: -393 steps per day, 95% CI: -735 to -52, $p = 0.026$). This suggests that while intervention participants achieved higher absolute step counts during Week 1, they experienced greater declines relative to their early-week performance compared to control participants. Among the daily step count predictors, later days in the first week showed stronger predictive power, with Day 7 being the most significant predictor (coefficient: 0.373, 95% CI: 0.261 - 0.485, $p < 0.001$) and Day 3 also contributing meaningfully (coefficient: 0.246, 95% CI: 0.071 - 0.422, $p = 0.006$).

Model diagnostics ***(Images 1-2)*** suggest adequate fit despite some heteroscedasticity, some outliers, and heavy tails which may indicate some individuals having patterns deviating more than expected. The VIF analysis (Image 2) indicates some multicollinearity among predictors, but most variance inflation factors remaining within acceptable ranges less than 5. The finding that intervention group assignment negatively predicts Week 2 performance, after controlling for Week 1 activity, suggests an effect where early intervention success may not be sustainable. This highlights the importance of sustained engagement strategies, as participants who showed initial responsiveness to interventions appeared more vulnerable to activity decline in the second week compared to steady state control participants.

4.4 Scientific Question 4: Individual Variability

Random-effects estimates from the 14-day mixed model point to marked person-to-person diversity. The baseline (intercept) variance was 1.19 on the log scale, so a typical pair of participants differed by a factor of 3 in their starting step count (e.g., one at 2 000 steps per day, another at 6 000). The linear and quadratic day-slope variances were even larger, confirming that individuals followed very different trajectories—some continued to climb throughout the fortnight, others levelled off or declined. By contrast, the within-person residual variance was only 0.19, implying that a participant's steps on any single day fluctuated by about 50 % around their own fitted curve.

When we asked whether this variability depended on intervention arm, the answer was yes. Levene's test applied to the model residuals gave $F = 14.8$, $p = 3.8 \times 10$, and the box-plot (Figure) shows visibly wider spreads for the Standard and Tailored groups than for Control. In plain terms, the motivational messages not only raised the average number of steps but also widened the gap between "high" and "low" responders—some participants added thousands of steps while others changed very little.

Possible refinements. Reporting an intraclass correlation (ICC) or variance-partition coefficient would help readers grasp the share of total variability that lies between vs. within participants; plotting the distribution of random intercepts or showing a handful of subject-specific fitted curves could further illustrate how widely trajectories diverge.

Table 4: Average Variance Components for Random Effects

Group	Effect	Variance
Residual	Residual	0.188
id	Intercept (Baseline)	1.193
id	Day (Linear Slope)	92.466
id	Day (Quadratic Slope)	41.408

Overall Intraclass Correlation Coefficient (ICC): 0.864 MMeaning 86.4 % of total variance is due to stable individual differences Remaining 13.6 % is due to day-to-day fluctuations within individuals Assuming ## Average baseline step count: 3521 steps/day ## Individual 1 SD above average: 10499 steps/day (+ 6978 steps) ## Individual 1 SD below average: 1180 steps/day (-2341 steps) Linear slope variance: 92.47 - indicates large individual differences in linear trends over time Quadratic slope variance: 41.41 - indicates individual differences in acceleration/deceleration patterns

Method 1: Group-specific ICC calculation

Method 2: Levene's Test for homogeneity of variance Use residuals from the overall model fitted on first completed dataset Levene's Test for Homogeneity of Variance (center = median) Df F value Pr(>F)
group 2 14.824 3.844e-07 ***

The study population demonstrated substantial individual differences in daily step count patterns over the 14-day observation period. The overall intraclass correlation coefficient (ICC) of 0.864 indicates that 86.4% of the total variance in daily step counts was attributable to stable differences between individuals, while only 13.6% reflected day-to-day fluctuations within individuals. This high ICC demonstrates that participants maintained relatively consistent activity levels relative to their personal baseline throughout the study period.

Individual baseline step counts varied dramatically across participants, averaging 3,521 steps per day. The range of individual differences was striking: participants one standard deviation above the mean averaged 10,499 steps per day (6,978 steps above average), while those one standard deviation below averaged only 1,180 steps per day (2,341 steps below average). This created a 9,319-step range between individuals at the

±1 standard deviation boundaries, illustrating the profound heterogeneity in physical activity levels within the study population.

The magnitude of individual-level variability differed significantly between the three study arms, as confirmed by Levene's test for homogeneity of variance ($F = 14.824$, $p < 0.001$). While all groups maintained high levels of individual consistency, the Tailored intervention group showed the highest individual stability with an ICC of 0.851 (85.1% stable individual differences), compared to more modest levels in the Control group (ICC = 0.744, 74.4%) and Standard intervention group (ICC = 0.739, 73.9%).

Paradoxically, despite showing greater individual stability in relative rankings, both intervention groups exhibited increased absolute variability in their day-to-day fluctuations. The residual standard deviation, which captures daily variation around each individual's trajectory, was substantially higher in intervention groups. The Standard intervention group showed 79% greater residual variance than controls (variance ratio = 1.786), while the Tailored group demonstrated 107% greater residual variance (variance ratio = 2.068). This suggests that while intervention participants maintained their relative position within the group hierarchy, they experienced larger day-to-day swings in actual step counts.

The contrasting patterns between stable individual differences and increased daily fluctuations in intervention groups reveals important insights about intervention effects. The high overall ICC indicates that targeting interventions based on baseline activity levels would be highly effective, as individuals maintained consistent relative activity rankings throughout the study. However, the increased day-to-day variability in intervention groups suggests that behavioral change efforts may introduce instability in daily routines, even when overall individual trajectories remain predictable.

The Tailored intervention group's combination of highest individual stability (85.1% ICC) with greatest daily fluctuations (107% higher residual variance) indicates that personalized approaches may enhance consistency in relative performance while simultaneously introducing greater absolute variation in daily behavior. This pattern suggests successful individual targeting combined with implementation challenges that create day-to-day inconsistency in achieving step count goals.

5 Discussion

This study shows that a simple daily reminder can meaningfully increase short-term step counts: participants receiving either Standard or Tailored messages added, on average, 600–2 000 steps per day in the first week and maintained an advantage of roughly 1 000 steps per day across the fortnight. From a public-health perspective, these increments approach one additional kilometre of walking and, if sustained, could translate into clinically relevant reductions in cardiometabolic risk.

Contrary to expectation, tailoring did not yield an extra benefit over generic content. One possibility is that message timing—a constant across both arms—matters more than bespoke wording during the first two weeks of behaviour change. Alternatively, our tailoring algorithm, which relied on coarse baseline characteristics and recent step history, may have lacked the richness needed to deliver truly personalised motivation. Future work should evaluate more sophisticated, context-aware tailoring that incorporates mood, location or social cues.

The interventions also widened the distribution of outcomes: some individuals exceeded 10 000 steps per day while others barely moved. High random-slope variances and larger residual spreads in the messaging arms suggest heterogeneous engagement and underscore the limits of one-size-fits-all prompts. Identifying predictors of "high" versus "low" responders—such as baseline motivation, phone notification settings or occupational constraints—could inform adaptive interventions that escalate support when early gains are modest.

Several limitations temper these conclusions. The 14-day window precludes insight into long-term maintenance; missing data, though handled with multiple imputation under a Missing-at-Random assumption, could still bias results if non-wear relates to low activity; and the volunteer sample limits generalisability to older or less tech-savvy populations. Step counts were the only outcome; future studies should incorporate heart-rate or GPS data to differentiate intensity and context of movement.

In summary, well-timed generic reminders appear as effective as simple tailoring for short-term activity gains, but both approaches leave a sizeable subset of users untouched. Research should now focus on longer follow-up, richer personalisation, and adaptive delivery strategies that respond to early usage patterns to maximise population-level impact.

Cross validate predictions optional

6 Appendix

Table 5: Summary of missing 'Female' values by ID

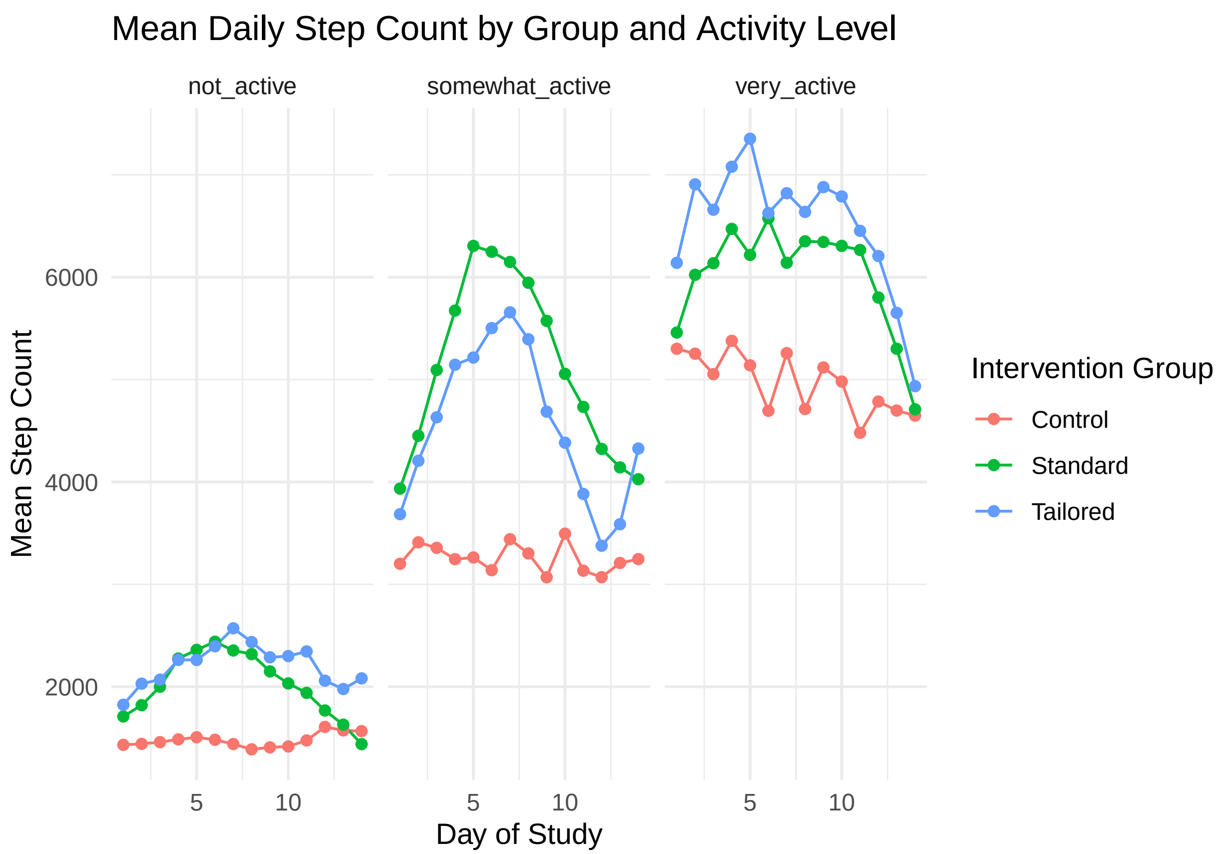
id	n_missing
38	14
206	14
237	14
252	14
281	14
283	14

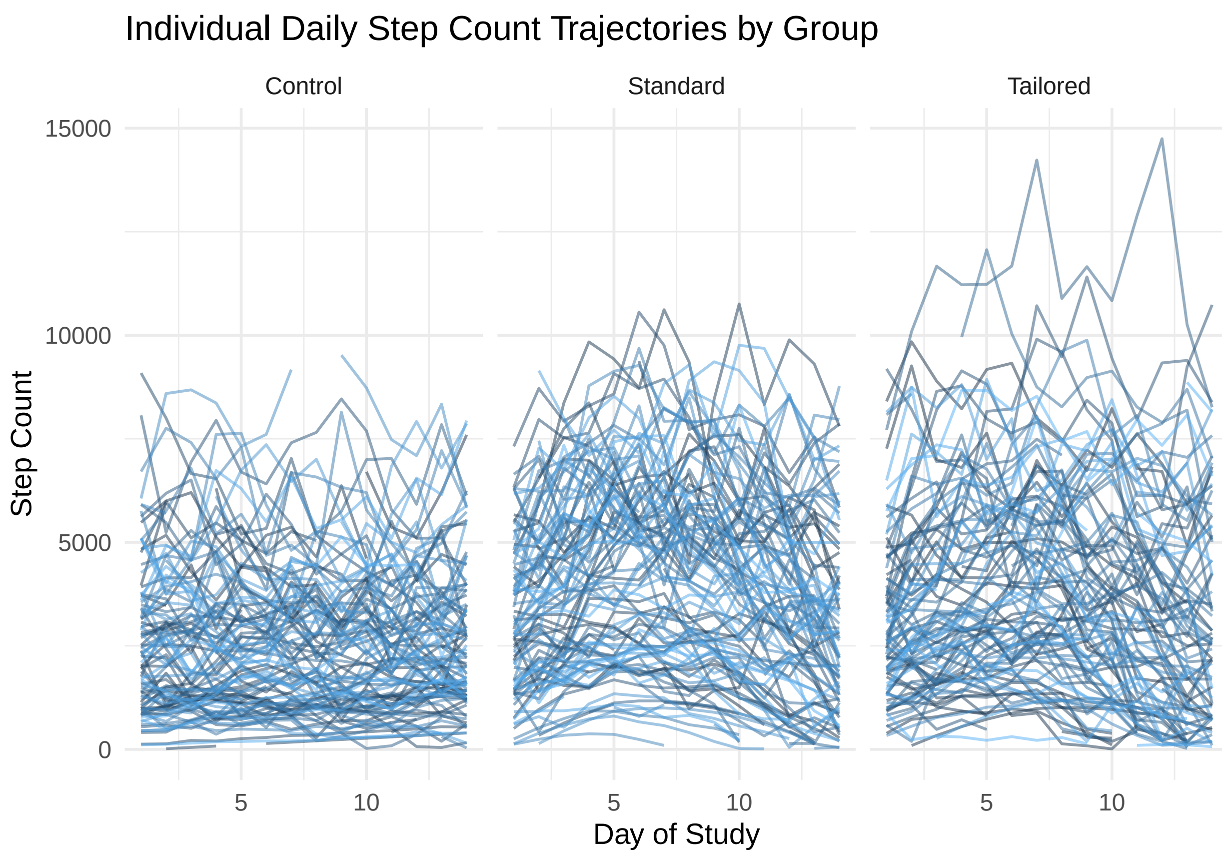
Table 6: Summary of Participants with Zero-Step or All-Missing Days

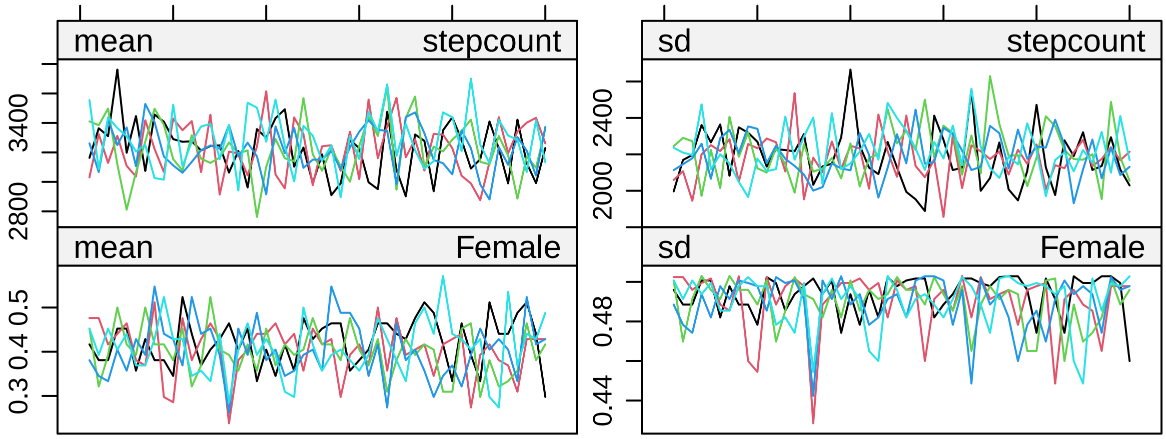
# 0 days	# of ids	Individuals ids
1	69	7, 9, 10, 13, 16, 19, 21, 24, 29, 31, 33, 36, 45, 53, 54, 55, 62, 64, 69, 75, 84, 85, 96, 101, 102, 105, 117, 122, 127, 130, 133, 136, 140, 142, 149, 151, 159, 160, 166, 168, 170, 179, 187, 198, 210, 211, 213, 218, 219, 220, 222, 230, 237, 239, 243, 247, 250, 251, 258, 263, 264, 267, 278, 287, 291, 297, 298, 299, 300
2	17	22, 27, 30, 32, 52, 76, 97, 123, 141, 145, 197, 215, 241, 252, 257, 269, 275
3	8	1, 112, 125, 174, 177, 178, 227, 273
4	7	17, 43, 44, 77, 147, 182, 207
6	1	225
9	1	93
14	2	47, 80

Table 7: Number of 0 Steps or N/A values

	Missing Count
id	0
day	0
dow	0
Group	0
Act	0
Age	0
Female	84
stepcount	170


Mean Daily Step Count by Group and Activity Level
not_active
somewhat_active
very_active
Mean Step Count
6000
4000
2000
5
10
Day of Study
Intervention Group
Control
Standard
Tailored


Individual Daily Step Count Trajectories by Group
Control
Standard
Tailored
15000
10000
5000
0
Step Count
5
10
Day of Study


mean
stepcount
sd
stepcount
mean
Female
sd
Female
2800
3400
2000
2400
0.3 0.4 0.5
0.44
0.48

Iteration

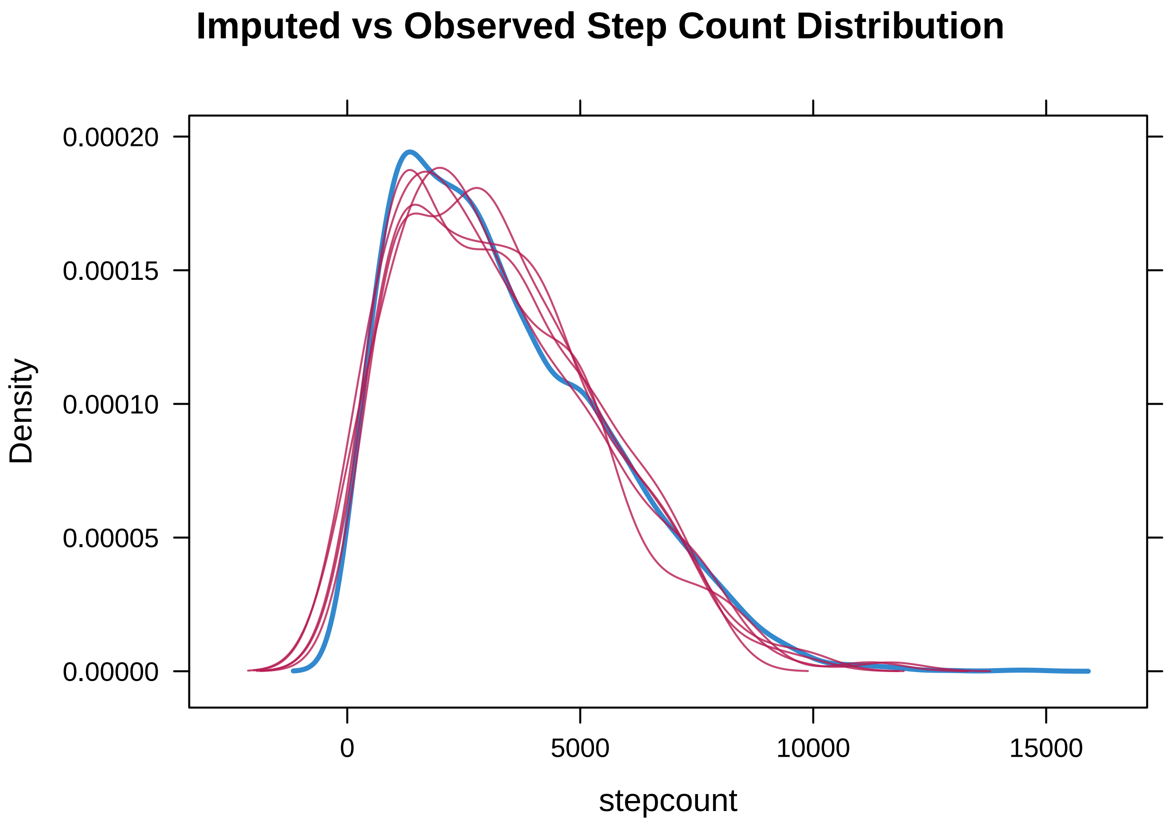
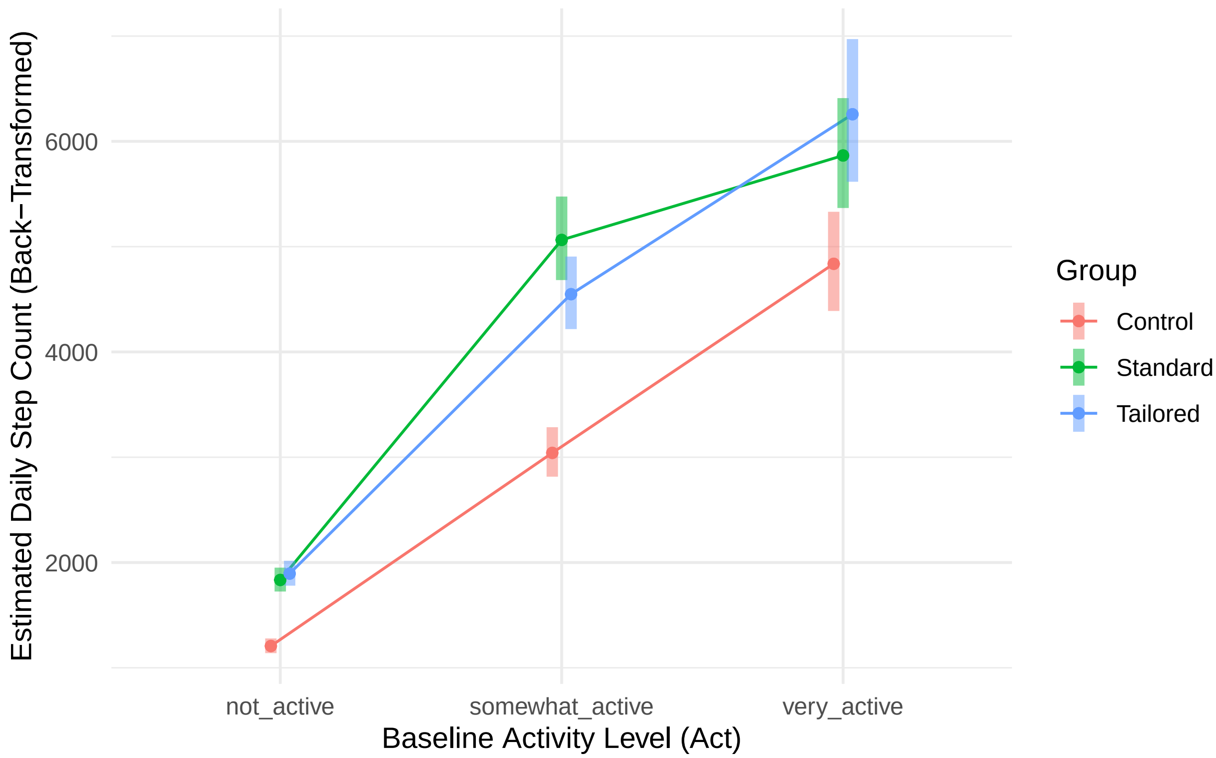
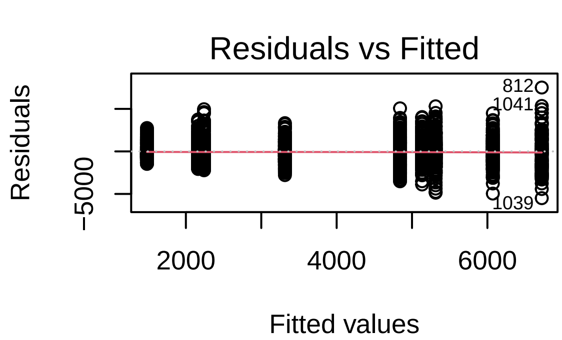
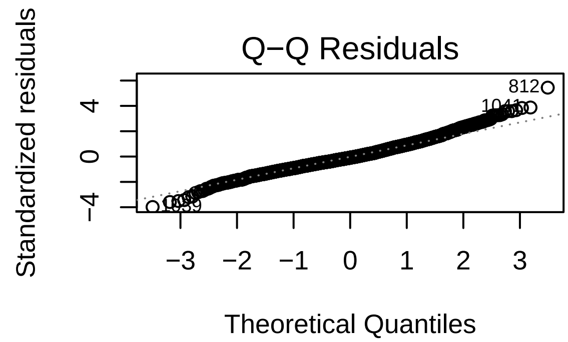
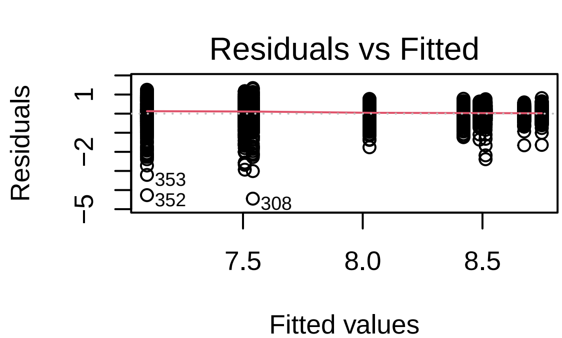
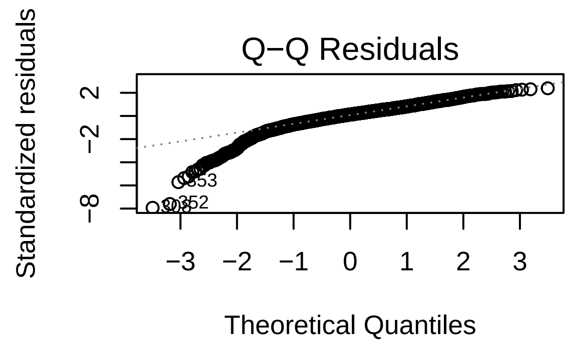
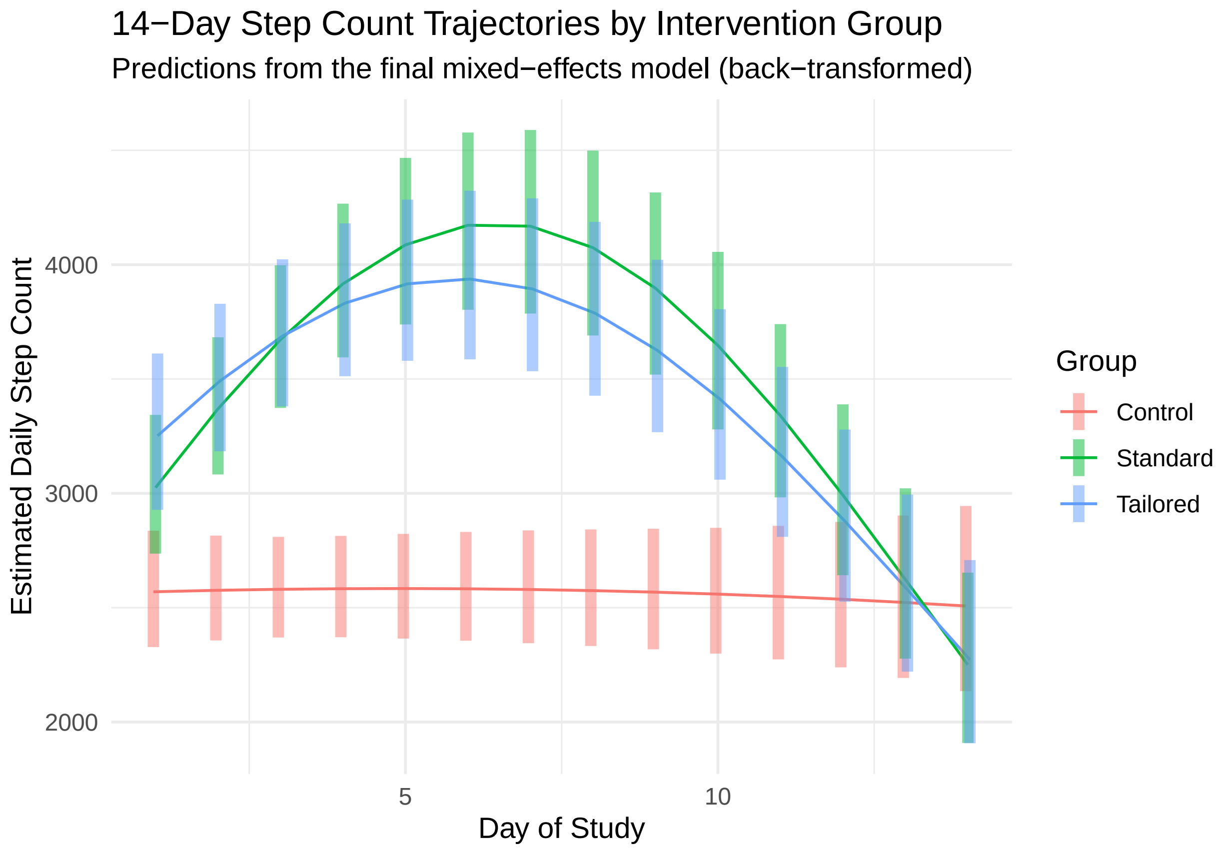

Imputed vs Observed Step Count Distribution
Density
0.00020
0.00015
0.00010
0.00005
0.00000
0
5000
10000
15000
stepcount

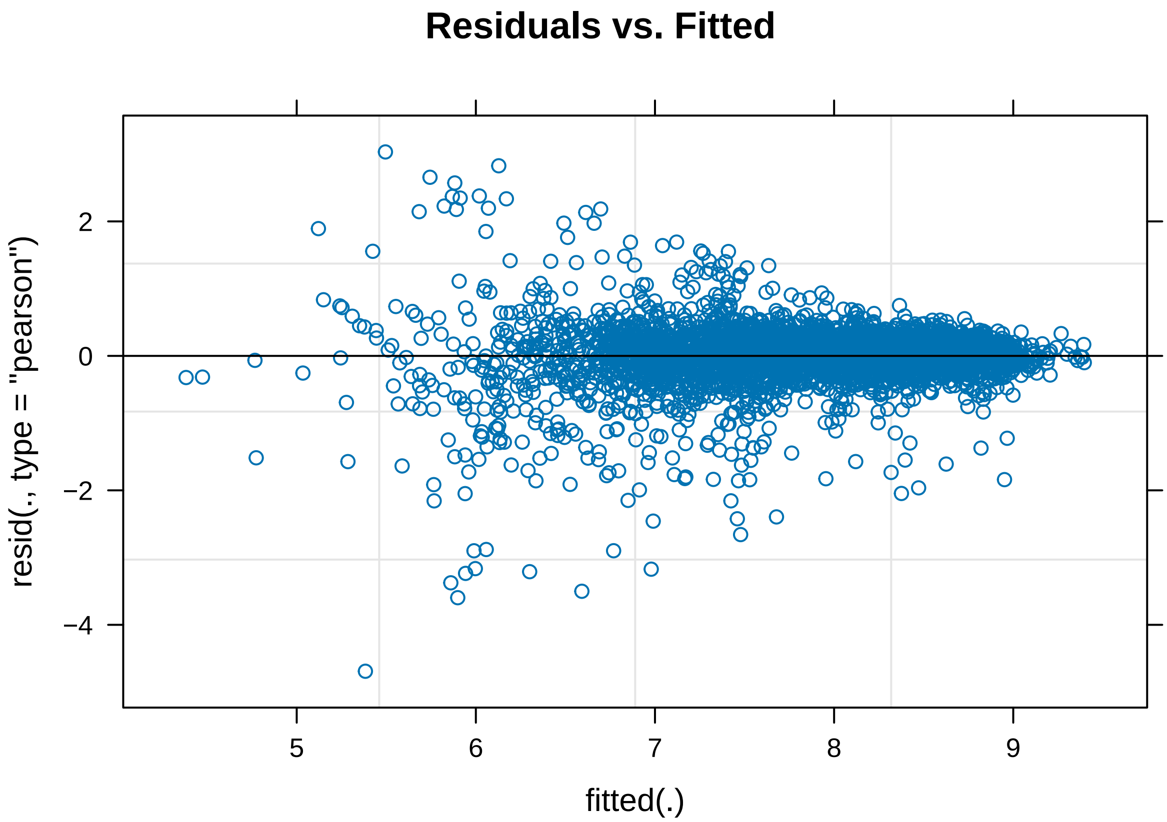
Figure 1: Interaction of Group and Baseline Activity on Week 1 Step Coun

Model estimates back−transformed from log scale




Residuals vs Fitted
Residuals
−5000
2000
4000
6000
Fitted values
812
1041
1039

Q−Q Residuals
Standardized residuals
4
0
−4
−3
−2
−1
0
1
2
3
Theoretical Quantiles
812

Residuals vs Fitted
Residuals
1
−2
−5
7.5
8.0
8.5
Fitted values
353
352
308

Q−Q Residuals
Standardized residuals
2
−2
−8
−3
−2
−1
0
1
2
3
Theoretical Quantiles


14−Day Step Count Trajectories by Intervention Group
Predictions from the final mixed−effects model (back−transformed)
Estimated Daily Step Count
4000
3000
2000
5
10
Day of Study
Group
Control
Standard
Tailored



Residuals vs. Fitted
resid(., type = "pearson")
fitted(.)
2
0
−2
−4
5
6
7
8
9

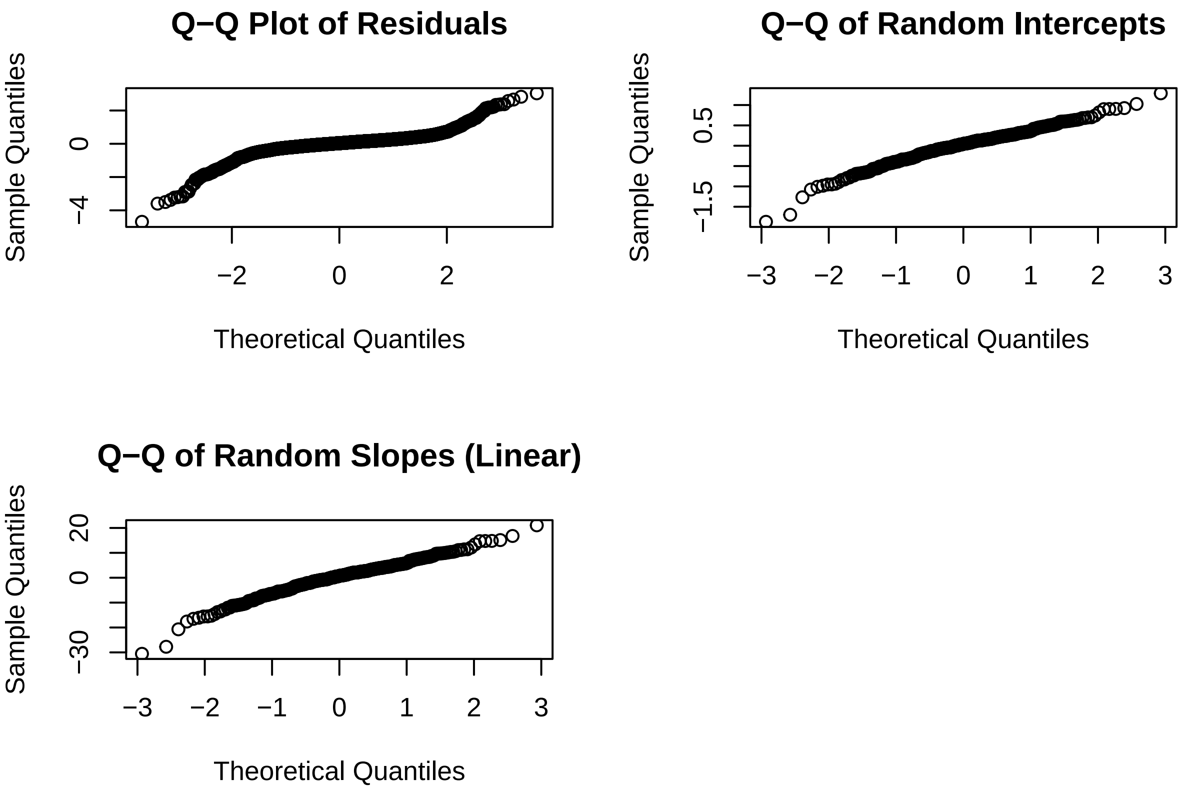
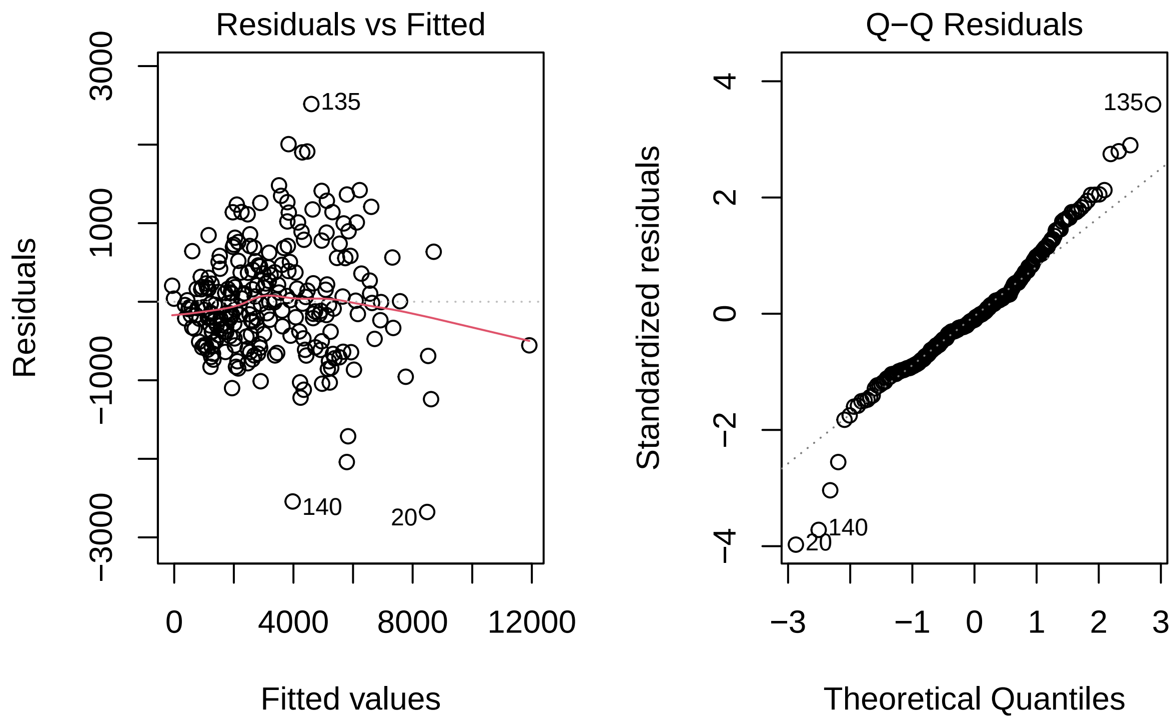


Table 8: Pooled Results for Multiple Linear Regression Model Predicting Week 2 Step Counts

Covariate	Estimate	95% Confidence Interval	P-Value
(Intercept)	673.623	-22.905 to 1370.151	0.058
GroupStandard	-375.658	-727.123 to -24.194	0.037
GroupTailored	-393.319	-735.025 to -51.612	0.026
Actsomewhat_active	-261.279	-587.035 to 64.476	0.115
Actvery_active	285.994	-192.978 to 764.967	0.240
Age	-5.950	-25.077 to 13.176	0.540
Female	-153.403	-355.233 to 48.427	0.136
day_1	-0.116	-0.257 to 0.026	0.108
day_2	-0.006	-0.179 to 0.167	0.947
day_3	0.246	0.071 to 0.422	0.006
day_4	0.046	-0.130 to 0.223	0.606
day_5	0.108	-0.082 to 0.299	0.263
day_6	0.181	-0.000 to 0.362	0.050
day_7	0.373	0.261 to 0.485	<0.001

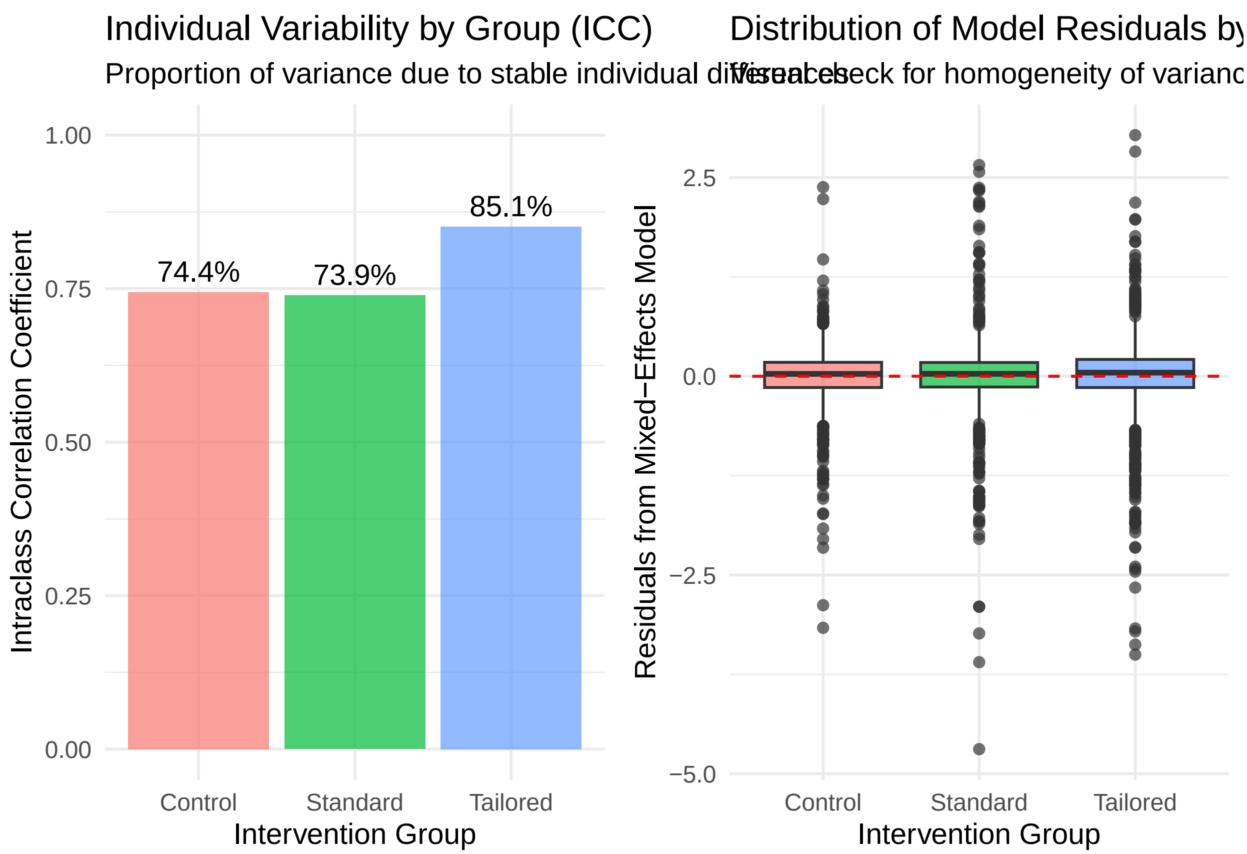


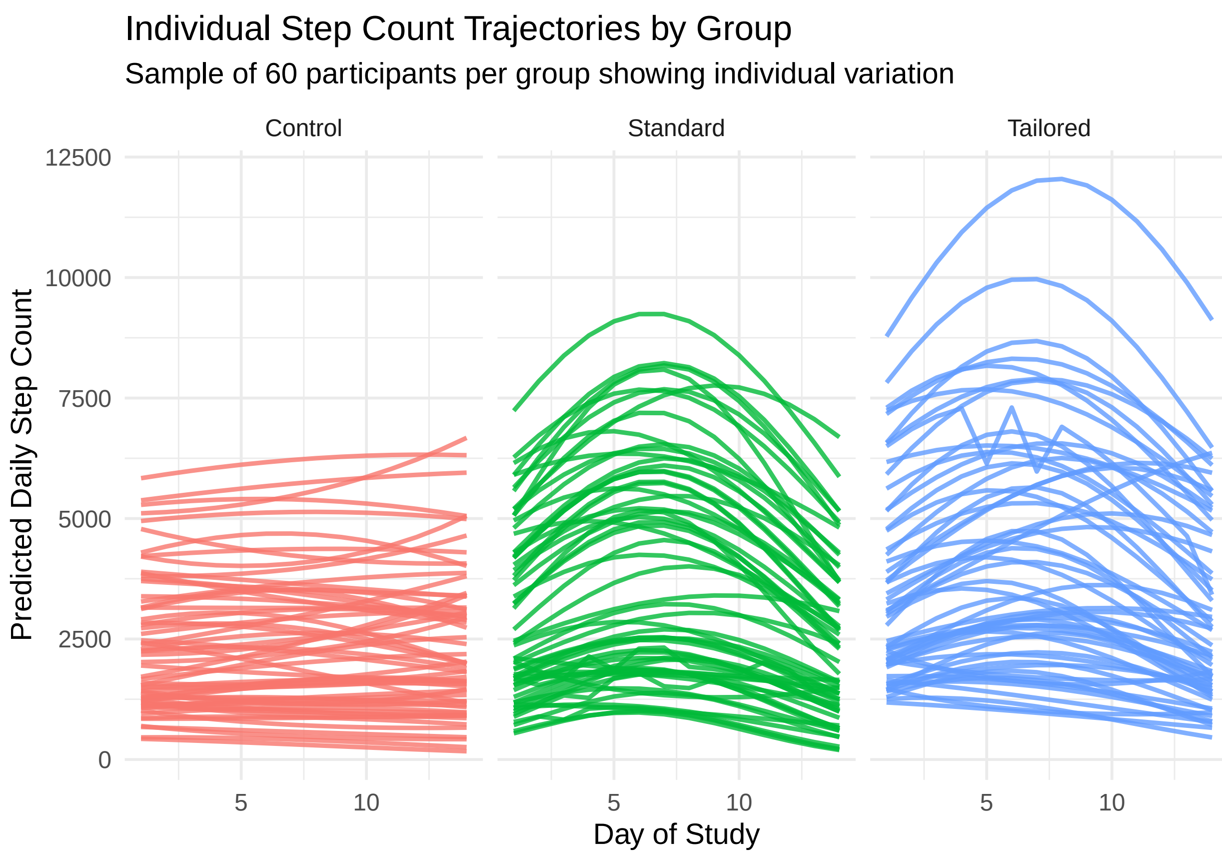

```
##              GVIF Df GVIF^(1/(2*Df))
## Group   1.518312  2        1.110044
## Act     4.352968  2        1.444430
## Age     1.254238  1        1.119928
## Female  1.082020  1        1.040202
## day_1   7.858439  1        2.803291
## day_2  12.684849  1        3.561580
## day_3  15.770379  1        3.971194
## day_4  18.436796  1        4.293809
## day_5  20.706658  1        4.550457
## day_6  16.444436  1        4.055174
## day_7   7.899663  1        2.810634
```

```
## Processing group: Control
## Processing group: Standard
```

```
## Processing group: Tailored
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##         Df F value    Pr(>F)
## group    2  14.824 3.844e-07 ***
##       4169
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


Individual Variability by Group (ICC)
Proportion of variance due to stable individual differences
Intraclass Correlation Coefficient
1.00
0.75
0.50
0.25
0.00
74.4%
73.9%
85.1%
Control
Standard
Tailored
Intervention Group
Distribution of Model Residuals by
Visual check for homogeneity of varianc
Residuals from Mixed−Effects Model
2.5
0.0
−2.5
−5.0
Control
Standard
Tailored
Intervention Group



Individual Step Count Trajectories by Group
Sample of 60 participants per group showing individual variation
Control
Standard
Tailored
Predicted Daily Step Count
12500
10000
7500
5000
2500
0
5
10
Day of Study

7 References

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